Credit-Risk Early Signals in Consumer Complaints

Final Experimental Report & Post–Checkpoint 2 Enhancements

1. Introduction

This project investigates whether **consumer complaint narratives** submitted to the U.S. Consumer Financial Protection Bureau (CFPB) contain **early warning signals** of credit-risk or dispute-prone behavior. Each complaint is labeled as “risky” if the consumer **disputed** the outcome or the company provided an **untimely response**. The core task is framed as a **binary text classification problem**, where the goal is to assign a risk score to each narrative.

The work has two main objectives:

1. **Predictive performance:** Build models that can reliably distinguish risky vs non-risky complaints under realistic temporal deployment conditions and severe **class imbalance (~2% positives)**.
2. **Interpretability and robustness:** Understand **which linguistic and contextual patterns** drive predictions, and evaluate **how well these models generalize** across different product domains (e.g., credit reporting vs debt collection).

Rather than predicting confirmed fraud or default events directly, this project focuses on identifying **early signals of potential credit or servicing risk embedded in consumer complaint narratives**. Complaint text often captures emerging issues before they appear in traditional financial datasets. By combining natural language processing with structured complaint metadata, the model aims to support **risk triage and regulatory monitoring workflows**.

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2. Data & Label Construction

2.1 Source and Sharding

The data comes from the **CFPB Consumer Complaint Database**, a large public repository of financial complaints. For computational efficiency:

- The raw CSV in Google Drive (≈11M complaints) was **stream-sharded** into 5 partitions.
- All subsequent experiments used **Shard 0** (≈2.3M rows), which was created by:
    - Reading the full file in chunks,

    - Assigning each row to a shard by `(row_index % N_SHARDS)`,
    - Writing only the rows with `row_index % 5 == 0`.

This preserved global diversity while keeping the working data manageable in Colab.

The CFPB complaint dataset contains millions of consumer complaints submitted against financial institutions. Each complaint includes structured attributes such as product category, company name, and state, as well as an unstructured narrative describing the consumer's issue.

2.2 Columns and Normalization

We selected the following columns:

- `date_received`
- `product`
- `consumer_complaint_narrative`
- `company`
- `state`
- `timely_response`
- `consumer_dispute`

Attribute	Description
Complaint narrative	Consumer description of issue
Product	Financial product category
Company	Financial institution
State	Consumer state
Date received	Complaint submission date
Consumer disputed?	Whether consumer disputed response
Timely response?	Whether company responded on time

Column names were normalized to `snake_case` (lowercase, underscores, and no punctuation) for consistency.

2.3 Text Cleaning and Filtering

We applied a **conservative but enhanced text normalization** pipeline:

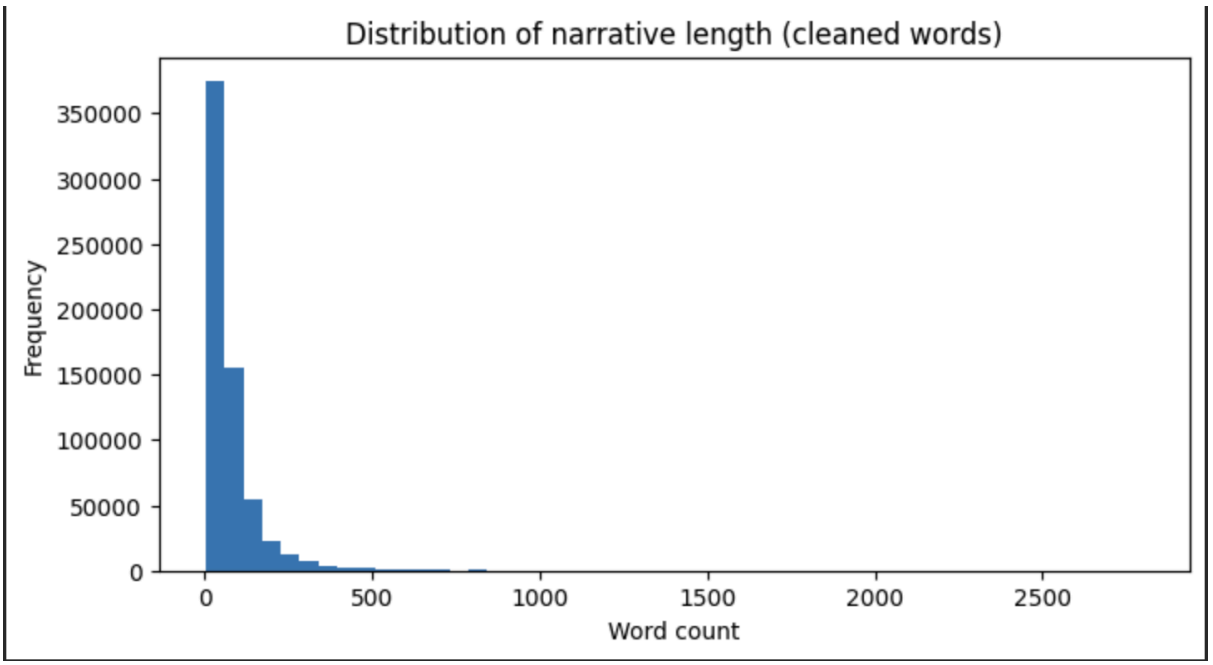
- Lowercasing the narrative.
- Removing URLs, emails, and obvious PII tokens (phone numbers).

- Removing non-alphabetic characters and normalizing whitespace.
- Optionally incorporating:
    - **Lemmatization** (NLTK WordNet) and
    - **Domain-specific stopwords** (e.g., boilerplate phrases that add little signal).

We then:

- Kept only rows with **non-empty narratives**.
- Filtered out narratives with **fewer than 5 tokens**.

Filtering very short narratives ensures that the model learns from complaints containing meaningful linguistic signals rather than noise or empty text.



- After cleaning and filtering, the final modeling dataset contained **679,808** complaints. The average raw narrative length was **1003.43** characters, and the average cleaned narrative length was **74.46** words. The retained dataset still spans **21 product categories, 4,336 companies, and 62 state values**, indicating that the filtering process removed low-information records while preserving broad product and institution coverage for downstream modeling.

2.4 Label Definition (`risk_flag`)

We constructed a **proxy risk label** using distant supervision from structured fields:

- `risk_flag = 1` if:
    - `consumer_disputed == "Yes"` **OR**
    - `timely_response == "No"`
- `risk_flag = 0` otherwise.

Before labeling, `consumer_disputed` and `timely_response` were normalized (trimmed, title-cased, irregular tokens mapped to NaN).

Label distribution on the cleaned subset:

- Non-risky (`0`): **665,713** (~97.93%)
- Risky (`1`): **14,095** (~2.07%)

This confirms that the task is **highly imbalanced and motivates the use of rare-event metrics** such as precision-recall AUC, recall-weighted F2, threshold optimization, and calibration analysis rather than raw accuracy.

The label used in this study represents a proxy supervisory risk signal rather than confirmed fraud or default. Complaints that are disputed by consumers or that receive untimely responses may indicate unresolved issues or potential regulatory concerns.

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3. Train/Test Protocol (Temporal Split)

To mirror realistic deployment, we used a **temporal split**:

- **Train set**: complaints with `date_received` < 2025-07-01
- **Test set**: complaints with `date_received` ≥ 2025-07-01

This avoids **temporal leakage** (e.g., learning from future language patterns or regulatory changes).

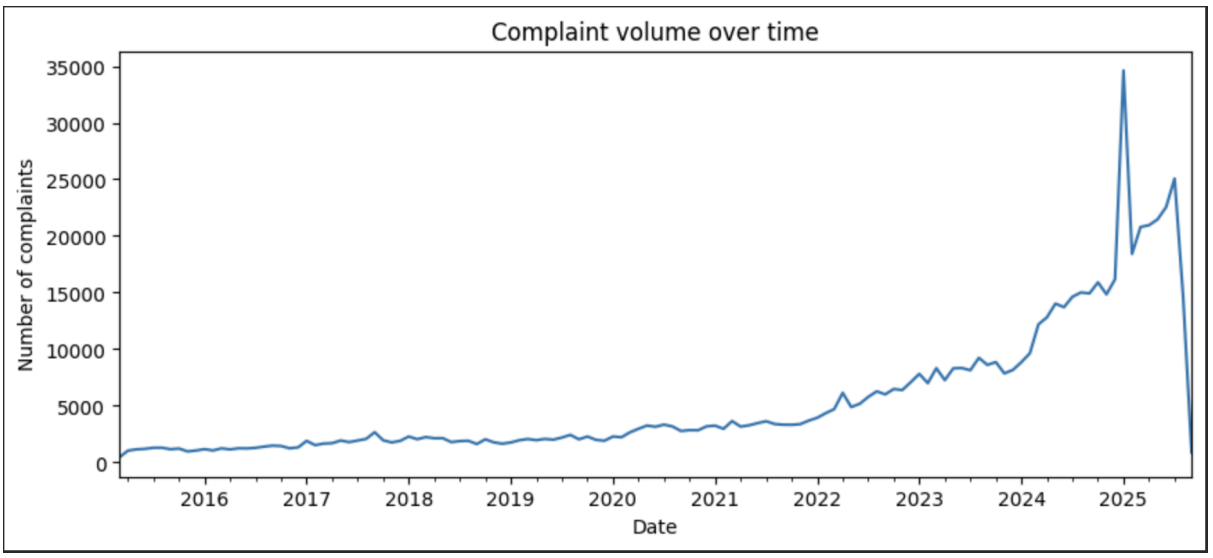
Final temporal split (after cleaning):

- Train size: 639,149 (positives ≈ 2.16%)
- Test size: 40,659 (positives ≈ 0.70%)

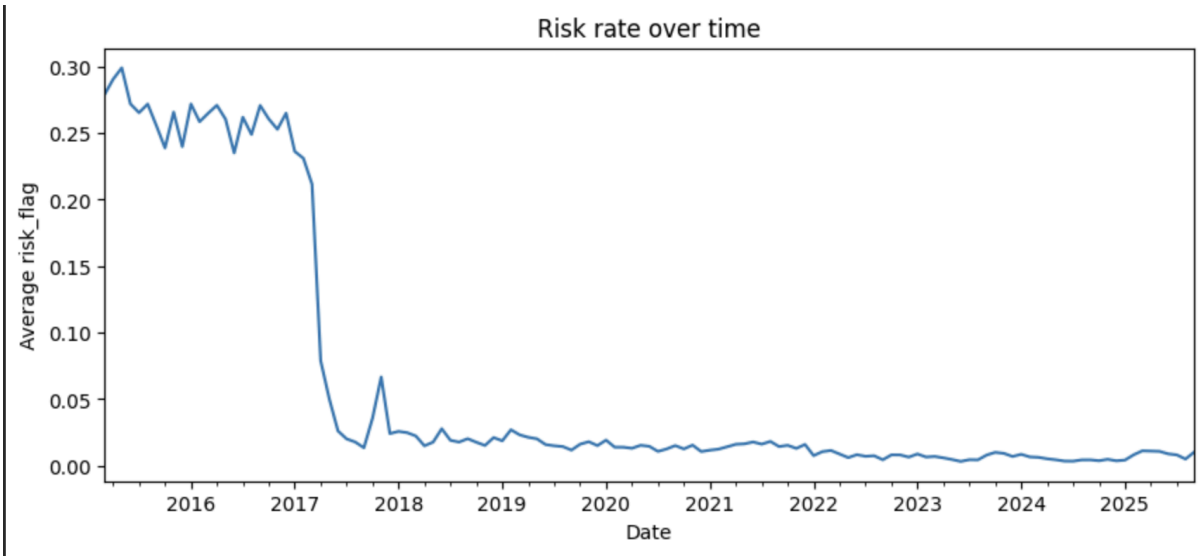
The lower positive rate in the test period makes the **precision–recall problem even harder** and gives a realistic stress test for an early-warning system.

Instead of randomly splitting the dataset, a temporal split was used to better simulate real-world deployment. Models were trained on complaints received before July 2025 and evaluated on complaints received afterward. This approach prevents information leakage and provides a realistic assessment of model performance on future data.

This temporal design is important because complaint language, complaint volume, and product mix evolve over time. A random split would overestimate performance by allowing the model to learn patterns that are too close to the future evaluation period. By contrast, the temporal split provides a more realistic estimate of how well an early-warning complaint model would generalize to newly arriving complaints.



Complaint volume fluctuates over time and shows increasing activity in recent years.



The risk rate declines over time, which partially explains the lower positive rate in the test set.

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4. Feature Engineering

We constructed a hybrid sparse feature space that combines lexical information from complaint narratives, structured complaint metadata, and engineered numeric/domain indicators. This design allows the models to use both what the consumer wrote and the institutional context in which the complaint occurred.

4.1 Word-Level TF-IDF Features

We used TfidfVectorizer with unigram and bigram features, ngram_range=(1,2), max_features=20,000, min_df=5, and max_df=0.9. This produced sparse text matrices with shape **(639,149 × 20,000) for training and (40,659 × 20,000) for testing**. These features capture topical vocabulary and short phrases while limiting vocabulary growth.

4.2 Structured Metadata Features

We one-hot encoded three structured metadata fields: **product, company, and state**. This produced sparse structured matrices with shape (639,149 × 4,373) for training and (40,659 × 4,373) for testing. These features allow the model to incorporate product-specific and institution-specific priors that are often highly informative in complaint-risk detection.

4.3 Numeric and Domain Features

In addition to lexical and structured features, we added nine numeric/domain features: sent_neg, sent_neu, sent_pos, sent_compound, urgency_count, has_urgency_word, dispute_keyword_count, has_dispute_keyword, and clean_word_count. These variables summarize sentiment polarity, urgency language, dispute-related vocabulary, and complaint length, and they provide compact domain signals that complement the sparse text representation.

4.4 Final Hybrid Feature Matrix

The final modeling matrix was created by horizontally concatenating the TF-IDF text features, structured one-hot features, and numeric/domain features. This yielded X_train with shape (639,149 × 24,382) and X_test with shape (40,659 × 24,382). This hybrid representation serves as the primary feature space for the main classical models evaluated in the report.

	feature_group	dimension
0	TF-IDF text	20000
1	Structured metadata	4373
2	Numeric/domain features	9

5. Models and Methods

We evaluated four model families to understand the tradeoff between predictive performance, interpretability, and representational richness.

5.1 Hybrid Sparse Models

The main classical models were trained on the full hybrid feature space described in Section 4. The first was **Logistic Regression** with max_iter=200 and class_weight="balanced". The second was **LightGBM** with n_estimators=200, learning_rate=0.05, scale_pos_weight set to the observed negative-to-positive class ratio in the training data, and random_state=42. The class imbalance ratio in the training set was 45.29.

5.2 Dense Embedding Models

To test whether dense semantic representations could outperform sparse lexical features, we encoded complaint narratives using **sentence-transformers/all-MiniLM-L6-v2.** Because embedding generation is computationally expensive, embeddings were computed for a 200,000-row training subset and the full temporal test set. The resulting matrices had shape (200,000 × 384) for training and (40,659 × 384) for testing. These 384-dimensional vectors were then used with Logistic Regression and LightGBM.

5.3 Fine-Tuned Transformer Classifier

We also evaluated an end-to-end **fine-tuned transformer using distilbert-base-uncased.** Because of severe class imbalance and notebook compute limits, the transformer was trained on a balanced subset of 27,616 complaints created by downsampling the negative class to match the number of positives. The resulting transformer training set had a positive rate of 0.50. The model was trained for two epochs and then evaluated on the full temporal test set.

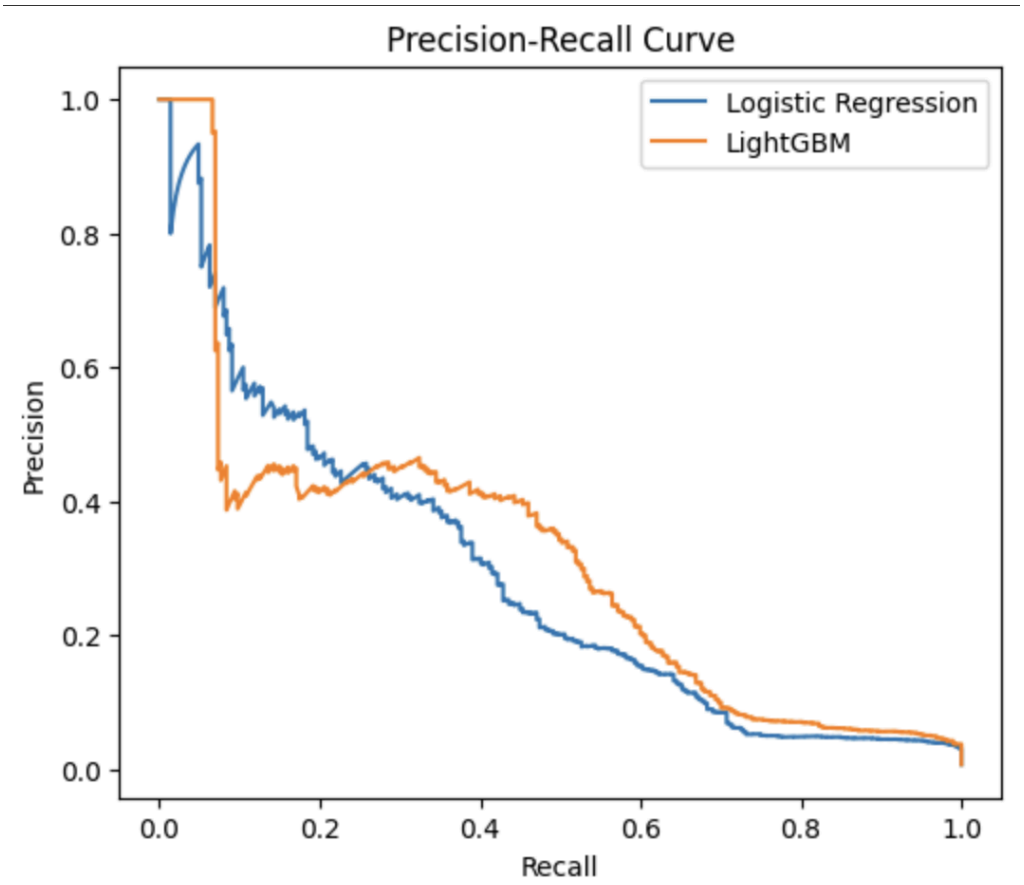
5.4 Cross-Product Generalization

Finally, we ran a domain-shift experiment in which the model was trained on one complaint product domain and evaluated on another. This experiment tests whether complaint-risk signals transfer cleanly across product categories or remain strongly product-specific.

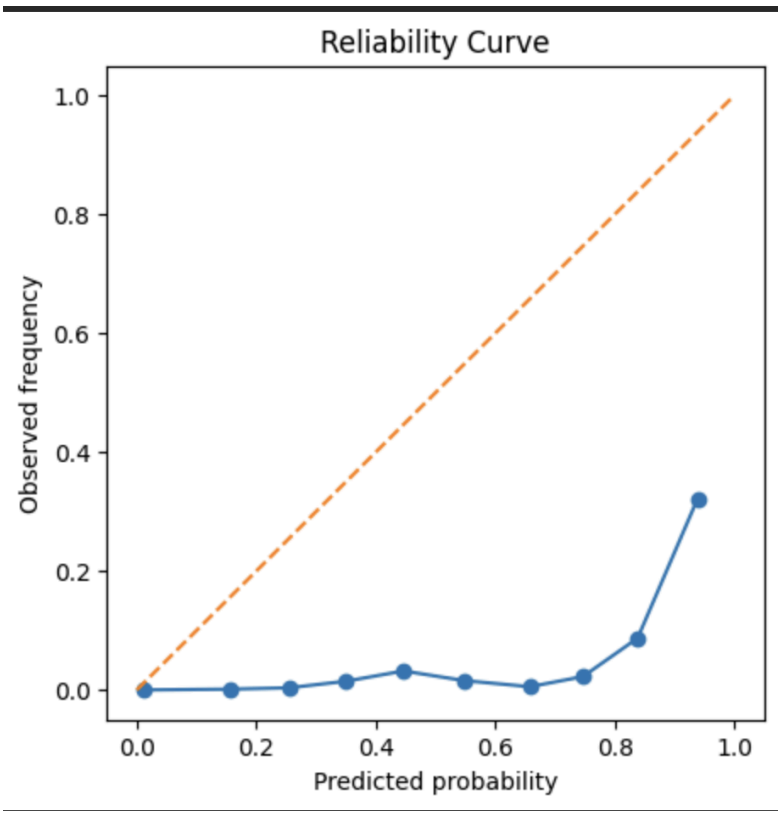
6. Evaluation Metrics

1. Because the temporal test set contains only 287 positive examples out of 40,659 complaints, accuracy is not an informative metric for this problem. Instead, the evaluation emphasizes AUROC for ranking quality, PR-AUC for rare-event retrieval performance, F2 for recall-weighted operating-point selection, and calibration metrics for probability quality.

2. For the main hybrid models, we generated precision-recall curves and selected the threshold that maximized F2 on the temporal test set. For the best hybrid model, the F2-optimal threshold was 0.801919, yielding a best F2 score of 0.390290. We also performed cost-based threshold analysis, which selected a lower operating threshold of 0.346566 and produced a minimum expected cost of 75,080 under the notebook's cost assumptions.



3. Because risk triage systems often use predicted probabilities directly, we additionally evaluated calibration. The uncalibrated model achieved a Brier score of 0.068151, and post-calibration this improved to 0.006271. This indicates that calibration materially improves the reliability of predicted probabilities, even when ranking performance is already strong.



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7. Results

7.1 Hybrid Sparse Models

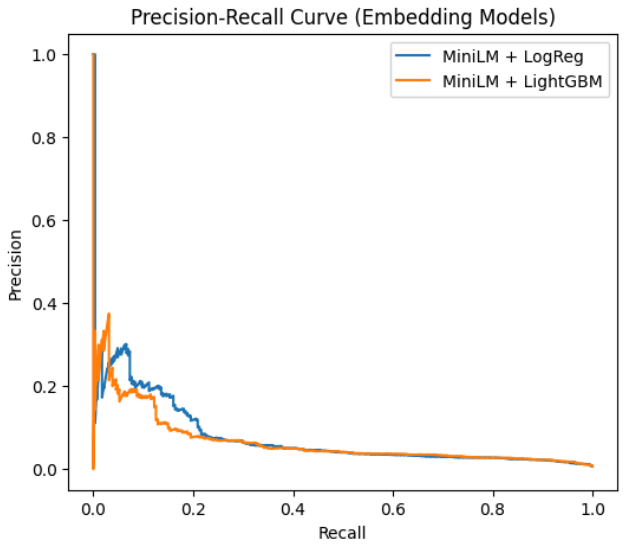
The strongest results came from the hybrid sparse feature space that combined TF-IDF text features, structured metadata, and numeric/domain signals. Logistic Regression achieved an AUROC of 0.955491 and a PR-AUC of 0.286208. LightGBM performed even better, achieving an AUROC of 0.966162 and a PR-AUC of 0.313761. These results establish LightGBM as the best predictive model in the current notebook.

The improvement over text-only modeling is substantial. In this problem, structured institutional context and compact domain signals meaningfully complement complaint language. The hybrid setup therefore provides a strong deployment-oriented representation for complaint-risk screening.

7.2 Dense Embedding Models

Dense MiniLM sentence embeddings underperformed the hybrid sparse models. Logistic Regression on 384-dimensional MiniLM embeddings achieved an AUROC of 0.874315 and a PR-AUC of 0.076682, while LightGBM on the same embeddings achieved an AUROC of

0.880024 and a PR-AUC of 0.066838. The best embedding model was MiniLM plus Logistic Regression, with a best F2 threshold of 0.841515 and a best F2 score of 0.177410.



7.3 Fine-Tuned Transformer Classifier

The fine-tuned transformer experiment used distilbert-base-uncased rather than MiniLM. Training was performed on a balanced subset of 27,616 complaints with a 50% positive rate. On the full temporal test set, the fine-tuned transformer achieved an AUROC of 0.904745 and a PR-AUC of 0.120720. This clearly improves over static embedding baselines, but it still does not match the best hybrid sparse models, likely because the current notebook uses a smaller balanced subset for fine-tuning and the task benefits strongly from structured metadata.

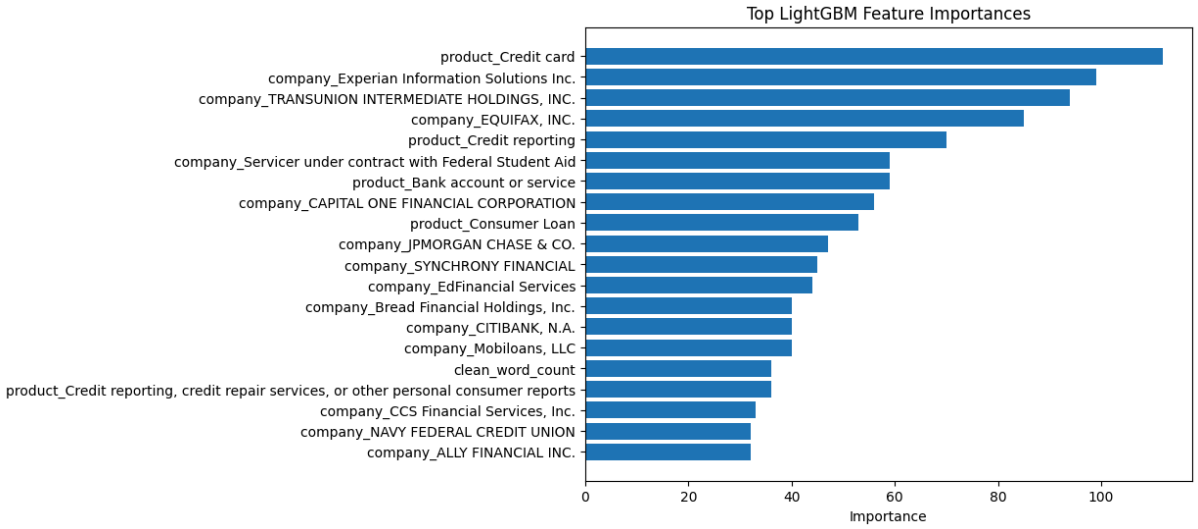
7.4 Cross-Product Generalization

To assess domain shift, we trained on complaints from the product category "Credit reporting, credit repair services, or other personal consumer reports" and evaluated transfer to "Debt collection." The source domain contained 160,385 complaints and the target domain contained 74,099 complaints. In this cross-product setting, the model achieved an AUROC of 0.720710 and a PR-AUC of 0.123574. These results are substantially weaker than the in-domain hybrid results, indicating that the learned signals are only partially portable across complaint product categories.

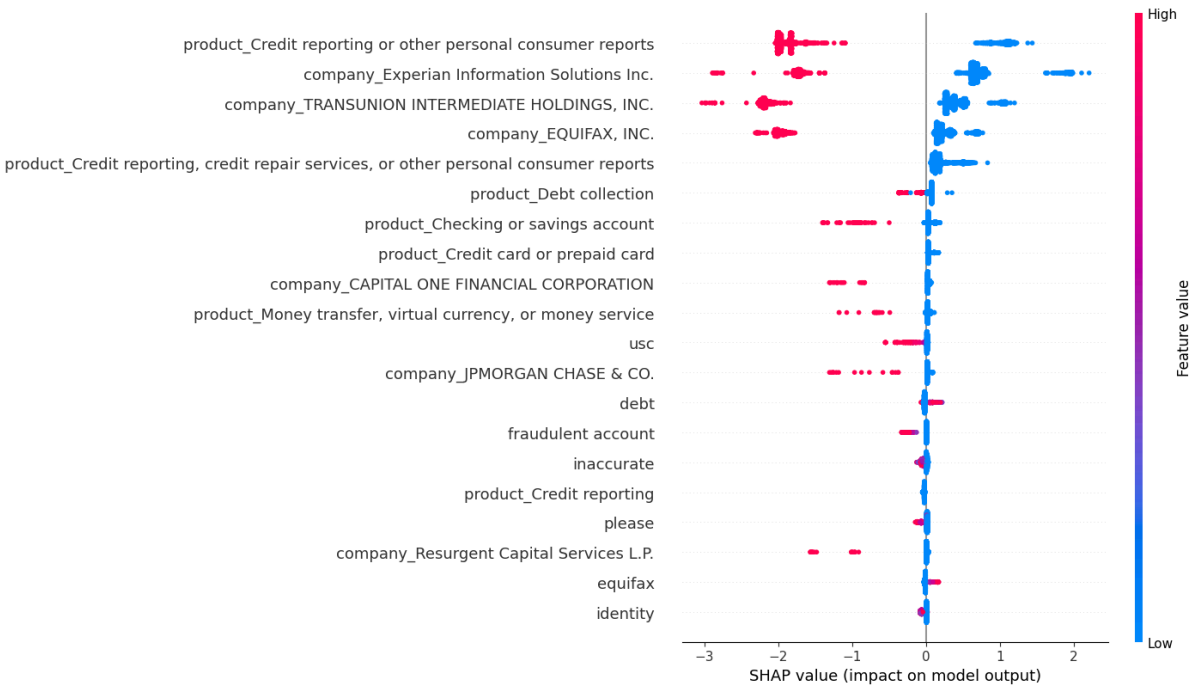
7.5 Interpretability Findings

The hybrid logistic regression model remains highly interpretable. Its strongest positive coefficients include product_Credit reporting, company_Servicer under contract with Federal Student Aid, product_Consumer Loan, product_Bank account or service, company_CCS Financial Services, Inc., company_ROBINHOOD MARKETS INC., loan, sent_neu, and call. Its strongest negative coefficients include company_Experian Information Solutions Inc., company_TRANSUNION INTERMEDIATE HOLDINGS, INC., company_EQUIFAX, INC., product_Vehicle loan or lease, and product_Checking or savings account. This shows that

the linear model relies on a combination of complaint vocabulary, institutional context, and engineered domain signals rather than text alone.



SHAP analysis of the LightGBM hybrid model confirms a similar pattern. Product and company one-hot features are dominant drivers of model output, while words associated with credit-report disputes and account correction requests also contribute strongly. Taken together, the coefficient analysis and SHAP results show that the model is learning plausible but highly product-specific risk patterns.



9. Best Method and Overall Conclusions

9.1 Best Model for This Task

Considering predictive performance and deployment relevance, the best model in the current notebook is LightGBM trained on the hybrid TF-IDF + structured metadata + numeric/domain feature space. It achieved an AUROC of 0.966162 and a PR-AUC of 0.313761, outperforming the hybrid logistic regression model as well as all dense embedding and transformer baselines. Its advantage likely comes from capturing nonlinear interactions among complaint vocabulary, institutional context, and compact engineered risk signals.

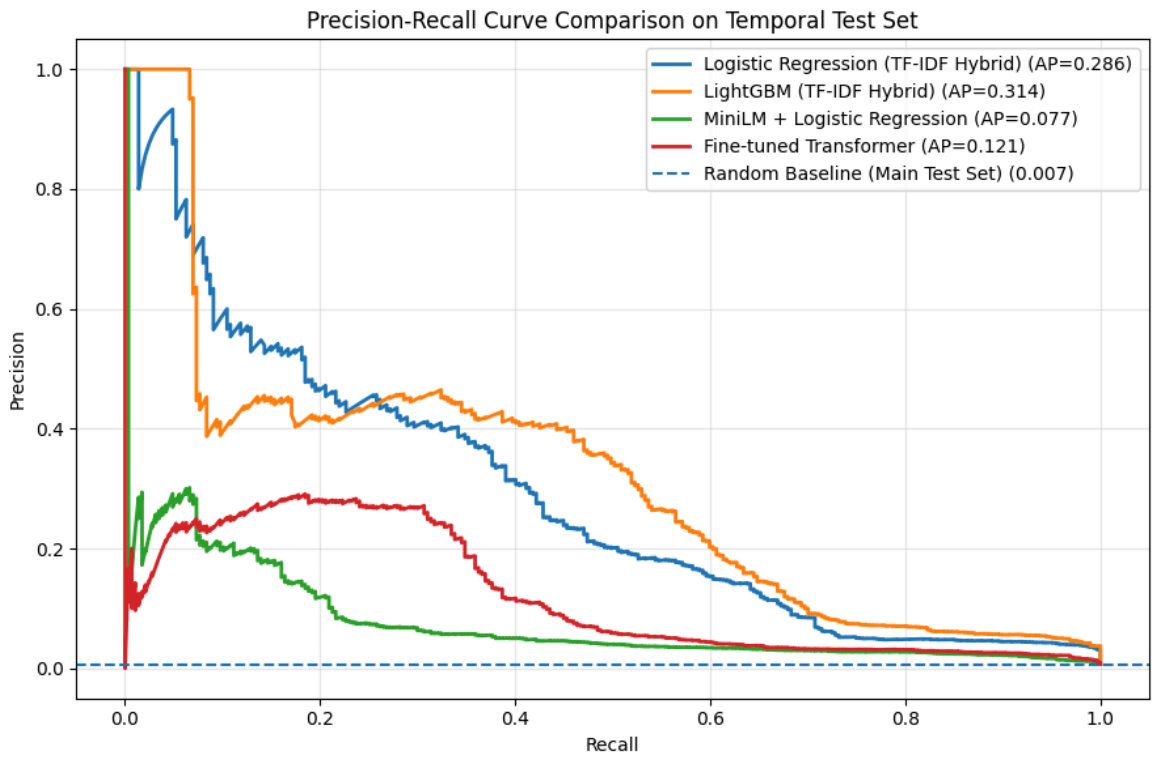
	Model	AUROC	PR_AUC
1	LightGBM (TF-IDF Hybrid)	0.966162	0.313761
0	Logistic Regression (TF-IDF Hybrid)	0.955491	0.286208
4	Fine-tuned Transformer	0.904745	0.120720
2	MiniLM + Logistic Regression	0.874315	0.076682
3	MiniLM + LightGBM	0.880024	0.066838

9.2 Role of Other Models

The hybrid logistic regression model remains extremely valuable because it offers strong performance together with direct interpretability. It achieved an AUROC of 0.955491 and a PR-AUC of 0.286208, making it a strong benchmark and a good candidate when transparency is a primary requirement.

The MiniLM embedding models demonstrate that dense semantic vectors alone are not sufficient in this task. They underperform the sparse hybrid models by a wide margin, suggesting that structured complaint context and sparse lexical cues remain essential.

The fine-tuned DistilBERT model improves meaningfully over static embedding baselines, achieving an AUROC of 0.904745 and a PR-AUC of 0.120720, but it still trails the best hybrid classical models in the current compute-constrained setup.



9.3 High-Level Findings

First, consumer complaint narratives do contain early signals of elevated risk, since even the baseline models achieve nontrivial predictive performance on a future temporal holdout set. Second, structured metadata and engineered domain indicators materially improve rare-event detection beyond text alone. Third, the best-performing system in the current project is a hybrid classical model rather than a dense embedding or fine-tuned transformer approach. Fourth, interpretability analysis shows that complaint-risk prediction is strongly shaped by product category, company identity, and recurring complaint language. Finally, cross-product transfer remains limited, indicating that product-aware models or domain-adaptive approaches may be necessary for broader deployment.